

Guidelines of DISCIPLINE SPECIFIC ELECTIVE/ GE Semester VII/VIII
(NEP-UGCF 2022)

Natural Language Processing

DSE/GE

(Effective from Academic Year 2025-26)

Tentative Guidelines. Subject to approval of the committee

Units	Chapter	Reference	No. of Hours
Unit 1: Introduction and Basic Text Processing: Knowledge in Speech and Language Processing, NLP Applications, The problem of ambiguity, Regular Expressions, Text Normalization, Tokenization, Stemming, Lemmatization, Stop-word removal	Chapter 1:Natural Language Basics Natural Language Processing (from Page 62 to 68) Chapter 2: Python for Natural Language Processing Regular Expressions (from page 102 to 106) Chapter 3: Processing and Understanding Text Text Tokenization (from page 119 to 127 uptill Default Word Tokenizer) Text Correction (Stemming from page 148 to 154 uptill Removing Stopwords)	[2]	7
Unit 2 Formal Language Modeling: Byte Pair Encoding and Edit Distance, Unigrams, Bigrams, N-grams, Markov assumption, Maximum likelihood estimation, N-gram Language Models, Smoothing and Perplexity	Chapter 2:Words and Tokens 2.4 Subword Tokenization: Byte-Pair Encoding (uptill 2.4.2) 2.9 Minimum Edit Distance Chapter 3:N-gram Language Models 3.1 N-grams 3.3 Evaluating Language Models: Perplexity (introduction excluding subsections) 3.4 Sampling sentences from a language model 3.6.1 Laplace Smoothing 3.6.2 Add-k smoothing	[1]	8

Unit 3 Vector Semantics and Embedding: Words and Vectors, Bag-of-Words (BoW), TF-IDF, Learning Word Embeddings - Word2vec	Chapter 4: Feature Engineering for Text Representation Traditional Feature Engineering Models (from page 208 to 213 - uptill TF-IDF Model) Chapter 5: Embeddings 5.2 Vector Semantics: The Intuition 5.3 Simple count-based embeddings 5.4 Cosine for measuring similarity 5.5. Word2vec (excluding subsections)	[2] [1]	7
Unit 4 Sequence Labeling for Parts of Speech Tagging: Part-of-Speech Tagging, Named Entities and Named Entity Tagging/Recognition	Chapter 17: Sequence Labeling for Parts of Speech and Named Entities 17.2 Part-of-Speech Tagging 17.3 Named Entities and Named Entity Tagging	[1]	3
Unit 5: Deep Learning Models for NLP: Feedforward Neural Networks, RNN, Applications of Text Mining: Text classification, Sentiment Analysis	Chapter 6: Neural Networks 6.1 Units 6.2 The XOR problem 6.3 Feedforward Neural Networks 6.4 Feedforward networks for NLP: Classification 6.5 Embeddings as the input to neural net classifiers 6.6 Training Neural Nets Chapter 13 RNNs and LSTMs 13.1 Recurrent Neural Networks (excluding subsections) 13.2 RNNs as Language Models (excluding subsections)	[1]	8

Unit 6 : Large Language Models for NLP: LLMs and Transformers, Limitations of static embeddings, Contextual representation based LM vector embeddings (BERT, RoBERTa, GPT), Ethical and safety considerations with Language Models	Chapter 7: Large Language Models 7.1 Three architectures for language models 7.2 Conditional Generation of Text: The Intuition 7.3 Prompting 7.4 Generation and Sampling 7.5 Training Large Language Models (uptill 7.5.1) 7.7 Ethical and Safety Issues with Language Models Chapter 8 Transformers 8.1 Attention (excluding subsections) 8.2 Transformer Blocks 8.3 Parallelizing computation using a single matrix X 8.4 The input: embeddings for token and position 8.5 The Language Modeling Head Chapter 9 Masked Language Models 9.1 Bidirectional Transformer Encoders 9.3 Contextual Embeddings (uptill 9.3.1 (not including Word Sense Disambiguation))	[1]	12
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References

1. Daniel Jurafsky and James H. Martin: *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition*: 3rd Edition, Pearson, 2026 (Draft version available online at https://web.stanford.edu/~jurafsky/slp3/ed3book_jan26.pdf)
2. Dipanjan Sarkar. Text Analytics with Python: A Practitioner's Guide to Natural Language Processing. Second Edition. APress. 2019

3. Steven Bird, Ewan Klein, and Edward Loper . *Natural Language Processing with Python – Analyzing Text with the Natural Language Toolkit*, 2022 (Available online at <https://www.nltk.org/book/>)

Additional References

1. Yoav Goldberg. *A Primer on Neural Network Models for Natural Language Processing*, 2022 (Available online at <https://u.cs.biu.ac.il/~yogo/nnlp.pdf>)
2. Jacob Eisenstein. Introduction to natural language processing. MIT Press, 2019.
3. Christopher D. Manning and Hinrich Schütze. *Foundations of Statistical Natural Language Processing*. MIT Press, 1999.

Suggested Practical List:

Python Packages like Scikit (SKLearn), NLTK, spaCy, gensim, PyTorch, transformers (HuggingFace) etc. may be used for programming

1. Prepare/Pre-process a text corpus to make it more usable for NLP tasks using tokenization, conversion to lowercase, removal of punctuation, filtration of stop words, stemming and lemmatization.
2. Use regex patterns to extract the usernames from the email addresses, hashtags, dates, and phone numbers present in a given text.
3. List the most common words (with their frequency) in a given text excluding stopwords.
4. Create the TF-IDF (Term Frequency -Inverse Document Frequency) Matrix for the given set of text documents
5. Build a simple statistical language model: estimate unigram and bigram probabilities with add-one smoothing and compute the probability of given sentences.
6. Perform POS tagging in a given text file. Extract all the nouns present in the text. Create and print a dictionary with frequency of parts of speech present in the document.
7. Identify and print the named entities using Name Entity Recognition (NER) for a collection of news headlines.
8. Classify movie reviews as positive or negative from the IMDB movie dataset of 50K movie reviews. (Link for dataset: <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-review>)
9. Build and train a text classifier for the given data (using textblob or simpletransformers or keras library)
10. Generate text using a character-based model using an appropriate dataset. Given a sequence of characters from a given data (eg "Shakespear"), train a model to predict the next character in the sequence ("e").